

AMANDEEP SINGH

Node.js Backend Developer | AI Integration

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Portfolio: <https://iaman.space/>
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SUMMARY

MCA fresher specializing in backend development with hands-on experience building production-grade systems. Built ShodhKartaChecker (AI-powered plagiarism detection engine) using Node.js, FastAPI, Redis task queues, and Groq AI, achieving 40% faster API response times.

Skilled in microservices architecture, JWT authentication, RBAC, and REST API design. Looking to contribute to a backend-focused team and solve real-world engineering problems.

SKILLS

- Languages: JavaScript (Node.js), Python, Typescript
- Frameworks: Express.js, React.js, Nest.js
- Databases: MySQL, MongoDB, Redis, PostgreSQL
- Tools: Docker, Git, GitHub, Postman, Nginx, PM2
- Deployment: VPS, Vercel, Render
- Technologies: REST APIs, JWT Authentication, API Security, AI API Integration
- Concepts: Data Structures, DB Design, Error Handling, MVC Architecture, Backend Optimization, Asynchronous Processing

CERTIFICATION AND TRAINING

Full Stack Web Development Training – Department of Computer Science Punjabi University, Patiala

(MCA Industrial Training Program)

Duration: 1 Jan 2026 – 5 May 2026

- Learned and worked JavaScript, React.js, Node.js, Express.js, MongoDB, Python and Redis
- Developed backend applications and AI-integrated projects during training
- Built an AI-Powered Plagiarism Detection System with Redis-based task processing and Groq AI API integration

PROJECTS

ShodhKartaChecker (AI-Powered Plagiarism Detection System) [GitHub](#) | [Live Demo](#)

- Built a distributed plagiarism detection platform using Node.js, Express.js, FastAPI, Redis, and MongoDB Atlas
- Deployed frontend on Vercel, REST API services on Render, and Python NLP processing services on a VPS environment
- Designed a decoupled microservices architecture where Node.js handled API orchestration and FastAPI processed heavy document analysis workloads
- Implemented Redis-based asynchronous task queues to improve API responsiveness and

- background document preprocessing
- Configured Nginx reverse proxy and PM2 process management for production deployment and service reliability
- Integrated Groq AI APIs for plagiarism explanation and AI-generated similarity analysis
- Optimized document-processing workflows, reducing API response time by 40%

Scalable Quiz Platform (Personal Project) [GitHub](#)

- Developed RESTful APIs using Node.js and Express.js for a multi-user quiz system
- Implemented JWT-based authentication and role-based access control
- Designed and optimized MySQL database schema for better performance
- Handled errors efficiently using centralized middleware

Hugli Printing Press(Freelance Project) [GitHub](#) | [Live Demo](#)

- Built user-friendly UI using HTML, CSS, JavaScript, and React.js
- Worked directly on client requirements and website presentation design
- Integrated Node.js backend with WhatsApp API to enable direct client messaging from the contact page

EDUCATION

Master of Computer Applications (MCA)
Punjabi University – Patiala, India
Expected: May 2026